



Cover Page

SECR1033-02 COMPUTER ORGANIZATION AND ARCHITECTURE

GROUP PROJECT

2025/2026 Semester 2

SECTION 02

NAME	MATRIC NUMBER	SELECTED TOOLS	VIDEO LINK
MUHAMMAD FAIRUZ BIN HERMAN (PROJECT MANAGER)	A25CS0267	Window Suite CrystalDiskMark	https://drive.google.com/drive/folders/1r2e6YIOCq1MXUBgrdK3rJRRpzhikcXWx
NASAAIE BIN NORISKAMAR	A25CS0118	Window Suite PassMark Performance Test	https://drive.google.com/drive/folders/1WZCV6sK3xO_LpzU-2697RM906oVHEgy?usp=sharing
MUHAMMAD NAJMI SHAHMI BIN MOHD LATPI	A25CS0279	Window Suite Cinebench 2026	https://drive.google.com/file/d/1ZmOeruCV0m7C7zB6Hg0Z4yU3HAHwERUs/view?usp=drive_link
AKMAL RAFIQUE BIN AHMAD RAPHAIE	A25CS0181	Window Suite HWiNFO64	https://drive.google.com/file/d/1ZfrUFHoAF2GHfwLjHeEP LFhSBjVO9biv/view?usp=sharing

Lecturer Name : Dr. Rashidah binti Kadir

Table of Contents

Cover Page	1
Table of Contents	2
1.0 Introduction	3
1.1 Cinebench 2026: Specifications and Justifications	3
1.2 HwiNFO64: Specifications and Justifications	4
1.3 PassMark PerformanceTest : Specifications and Justifications	5
1.4 CrystalDiskMark: Specifications and Justifications	6
2.0 Tools Installation and Configuration	7
2.1 Installation and Configuration of Cinebench 2026	7
2.1.1 Installation Manual	7
2.1.2 Configuration Settings	7
2.1.3 Screenshots	8
2.2 Installation and Configuration of HWiNFO64	10
2.2.1 Installation Manual	10
2.2.2 Configuration Settings	11
2.2.3 Screenshots	12
2.3 Installation and Configuration of PassMark PerformanceTest	14
2.3.1 Installation Manual	14
2.3.2 Configuration Settings	14
2.4 Installation and Configuration of CrystalDiskMark	16
2.4.1 Installation Manual	16
2.4.2 Configuration Settings	16
3.0 Performance BenchMark Analysis and Discussions	18
3.1 Detailed Theoretical Explanation of Workload A and Workload B	18
a. Cinebench 2026	18
b. HWiNFO64	19
c. PassMark PerformanceTest	20
d. CrystalDiskMark	23
3.2 Flowcharts for Executing Workload A (Core Processing Metrics)	24
a. Cinebench 2026	24
b. HWiNFO64	26
c. PassMark PerformanceTest	27
d. CrystalDiskMark	29
a. Cinebench 2026	30
c. PassMark PerformanceTest	33
d. CrystalDiskMark	34
3.4 Hardware Performance Analysis and Diagnostic Discussions	35
a. Cinebench 2026	35
3.4.1.1 Workload A : Multi-Core Custom Loop (30 Minutes)	36
3.4.1.2 Workload B : Single-Core Execution Loop (10 Minutes)	37
b. HWiNFO64	39
3.4.2.1 Workload A : Sensors-Only Background Monitoring	39
3.4.2.2 Workload B : Core Utility and Temperature Tracking	41
3.4.3.1 Workload A : Standard CPU Instruction Suite	43
3.4.3.2 Workload B : Floating-Point Math and Vector Math Portions	44
d. CrystalDiskMark	46
4.0 Conclusion and Reflection	47
4.1 Conclusion	47
4.2 Reflection	48
4.2.1 Reflection by Muhammad Najmi Shahmi bin Mohd Latpi (Cinebench R23/2026)	48
4.2.2 Reflection by Akmal Rafique bin Mohd Raphaie (HWiNFO64)	48
4.2.3 Reflection by Nasaai bin Noriskamar (PassMark PerformanceTest)	49
4.2.4 Reflection by Muhammad Fairuz bin Herman (CrystalDiskMark)	49
5.0 References	50
Appendices	51

1.0 Introduction

This project focuses on evaluating a computer system's performance from an architectural perspective using four diagnostic tools: Cinebench, Geekbench, HWiNFO64, and CrystalDiskMark. The main objective is to run specific tests under Workload A and Workload B to observe how physical hardware components handle operational stress and data movement in real-time.

By tracking these benchmarks, the project analyzes low-level hardware behaviors such as memory bus controller overhead, Direct Memory Access (DMA), cache line limitations, and system bus latency. Additionally, it investigates processor metrics like core pipeline efficiency, memory cache coherence, and thermal throttling. Ultimately, this report connects theoretical concepts with real-world system data to evaluate hardware balance and performance limits.

1.1 Cinebench 2026: Specifications and Justifications

Cinebench 2026 is an industry-standard, cross-platform hardware evaluation tool developed by Maxon Computer GmbH. It measures a system's computational performance by rendering a complex, photorealistic 3D scene. Unlike generic synthetic testing platforms, it is built directly upon the core framework of professional animation software Cinema 4D and utilizes the production-grade Redshift rendering engine to place an intensive load on processor execution pipelines. (Maxon Computer GmbH. (2025). *Cinebench 2026* (Version 2026.1.0) [Computer software]. Maxon Computer. <https://apps.microsoft.com/detail/9pgzkjc81q7j>

Version Number : 2026.1.0

Link to Download : <https://apps.microsoft.com/detail/9pgzkjc81q7j>

Advantages : It feature separate testing routines for Multi-Core and Single-Core processing. This directly allowed us to compare Workload A (parallel execution) with Workload B (single-threaded constraint) to observe how instruction throughput behaves when hardware constraints change.

Disadvantages : The Redshift framework demands massive system resource overhead (requiring a minimum of 12 GB to 16 GB of system RAM), which limits the ability to run this exact benchmark across older or legacy low-tier computing platforms.

1.2 HwiNFO64: Specifications and Justifications

HWiNFO64 is a professional grade system information and diagnostic utility designed to provide a comprehensive breakdown of installed hardware components and real-time sensor monitoring across Microsoft Windows environments. Created by REALiX, this graphical and command line benchmark tool is commonly used by hardware optimization engineers and system analysts to diagnose microarchitectural constraints, pipeline bottlenecks and thread scheduling efficiency. It works by interfacing directly with digital thermal sensors (DTS), internal registers, and motherboard microcontrollers to continuously log highly granular metrics such as individual core temperatures, dynamic power consumption, thermal throttling events and clock frequency deltas. While standard operating system utilities only show generic loads, HWiNFO64 excels at tracking hardware behaviors under intense operational stressors, allowing users to isolate physical bottlenecks and measure how efficiently the software threads map onto the active CPU core topologies.

Version Number : 8.48(2026-05-26)

Link to Download : <https://dappcdn.com/download/utilities/hwinfo>

Advantages : The software operates with minimal system overhead, especially in "Sensors-Only" mode to ensure that real-time data collection doesn't skew performance results, and it lets users continuously log these granular technical metrics into highly flexible format extensions like CSV files for deep dive statistical analysis.

Disadvantages : The sheer density of the raw data feed can be visually overwhelming and complex to navigate for inexperienced users, and because it is strictly limited to physical component monitoring, its network diagnostics and software-level tracing capabilities are relatively basic.

1.3 PassMark PerformanceTest : Specifications and Justifications

PassMark PerformanceTest is an industry-standard, cross-platform hardware evaluation tool developed by PassMark Software Pty Ltd. It measures a system's computational performance by executing a comprehensive suite of isolated diagnostic tests across multiple subsystems, including the CPU, graphics, memory, and storage. Unlike generic synthetic testing platforms, it allows researchers to isolate specific execution paths such as integer math, floating-point math, sorting algorithms, and SIMD/vector math to place an intensive and measurable load directly on the processor's microarchitectural execution pipelines. (PassMark Software Pty Ltd. (2026). *PassMark PerformanceTest*[Computer software]. PassMark Software.)

Version Number : 11.1

Link to Download : [PassMark PerformanceTest - PC benchmark software](#)

Advantages :It allows sub-test isolation to be performed specifically. This tool separates Integer Math, Floating Point Math, and Sorting operations, facilitating analysis of the efficiency of individual ALU and FPU units.Besides,This tool is designed to run with minimal background processes interference from the operating system, ensuring that the throughput data captured is more accurate.

Disadvantages : The workloads generated by this software are synthetic (artificial intelligence simulations). This means that they reflect the processor's peak theoretical capability and not real-world mixed usage patterns.

1.4 CrystalDiskMark: Specifications and Justifications

CrystalDiskMark is a simple disk benchmark software that provides comprehensive benchmarking for sequential and random read/write speeds, which are essential for system optimization. It is an open-source disk drive benchmark tool for Microsoft Windows created by Crystal Dew World. Based on Microsoft's MIT-licensed Diskspd tool, this graphical benchmark is commonly used to test the performance of solid-state storage. It works by reading and writing through the filesystem in a volume-dependent way, generating read/write speeds in sequential and random positions with varying numbers of queues and threads. Solid-state drives tend to excel at random I/O because, unlike traditional hard drives, they do not need to physically seek for a specific position on a spinning disk to read or write data. (Wikipedia contributors. (2026, June 23). *Thread (computing)* - Wikipedia. [https://en.wikipedia.org/wiki/Thread_\(computing\)](https://en.wikipedia.org/wiki/Thread_(computing)))

Version Number : CrystalDiskMark v8.0.3

Link to Download : <https://crystalmark.info/en/software/crystaldiskmark/>

Advantages : The software is highly accurate and reliable when comparing storage performance across different drives and configurations. It has a very clean interface that makes it easy to use without complex setups. The testing cycles are fast, and the industry-standard results make it useful for both enthusiasts and professionals.

Disadvantages : The software forces heavy, continuous read and write loops, running it too much can cause temporary heat spikes and minor wear on SSD flash memory. It is also strictly limited to storage drives, so it cannot measure CPU performance or thermal throttling. Finally, background Windows updates can sometimes interfere with the test and cause the scores to fluctuate slightly.

2.0 Tools Installation and Configuration

2.1 Installation and Configuration of Cinebench 2026

2.1.1 Installation Manual

1. **Step 1:** Open the Microsoft Store application on your Windows device and navigate to the search bar located at the top of the screen.
2. **Step 2:** Type "Cinebench 2026" into the search bar, hit Enter, and click on the official application listing developed by Maxon Computer.
3. **Step 3:** Click the "Get" or "Install" button to automatically download and securely install the application packages directly onto your system.
4. **Step 4:** Once the installation is complete, launch Cinebench 2026 from your Windows Start Menu and click "Accept" on the initial End User License Agreement (EULA) pop-up screen.

2.1.2 Configuration Settings

- Workload A Configuration :

Test Type: CPU (Multi-Core) Parallel Execution Loop.

Minimum Test Duration: Set to 30 minutes (1800 seconds) via the *File and choose Preferences* menu.

Hardware Utilization: All 4 physical cores and 8 logical threads on the Intel Core i5-1035G4 are fully engaged simultaneously.

Target Metric Focus: Evaluates multi-threaded instruction scaling, maximum throughput under sustained multi-threaded parallelism, and long-term thermal limits.

- Workload B Configuration :

Test Type: CPU (Single Core) Serial Execution Loop.

Minimum Test Duration: Set to **10 minutes** (600 seconds) via the *File and choose Preferences* menu.

Hardware Utilization: Constrained to exactly 1 active core / 1 logical thread, leaving the remaining core resources at minimal utilization.

Target Metric Focus: Evaluates single-threaded efficiency, microarchitectural instruction execution rates, and peak clock frequency thresholds without thermal boundaries.

2.1.3 Screenshots

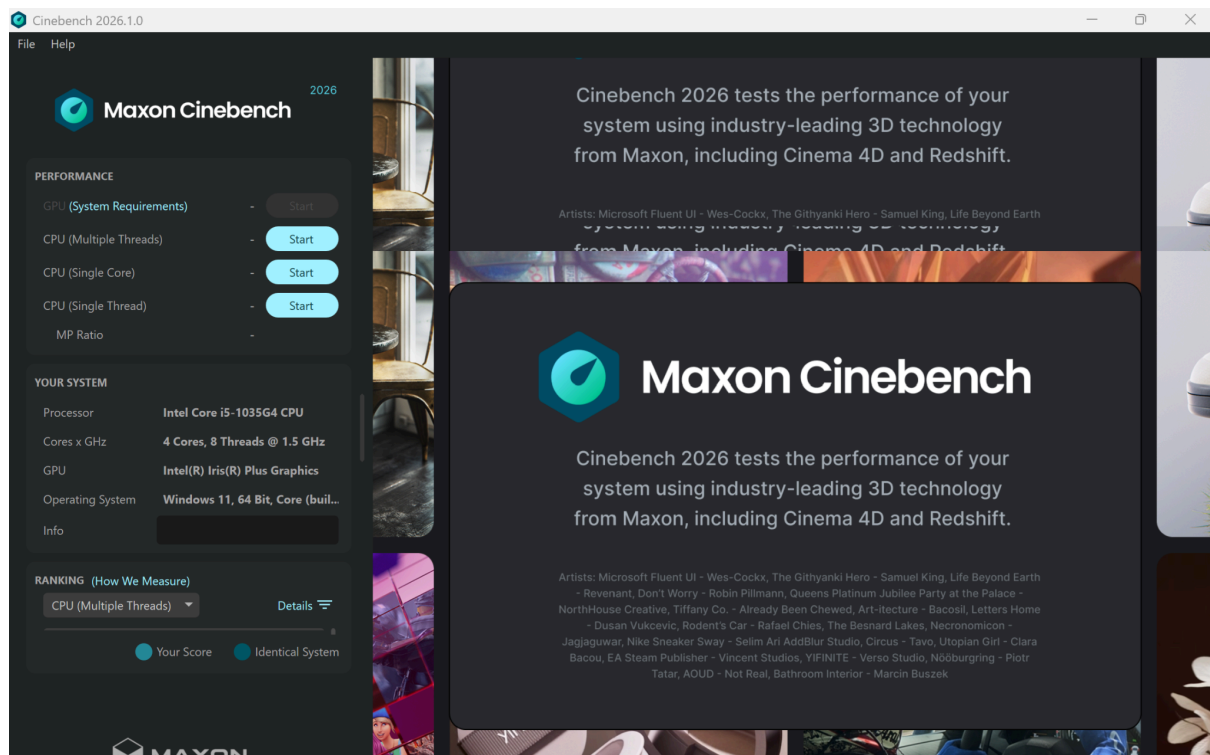
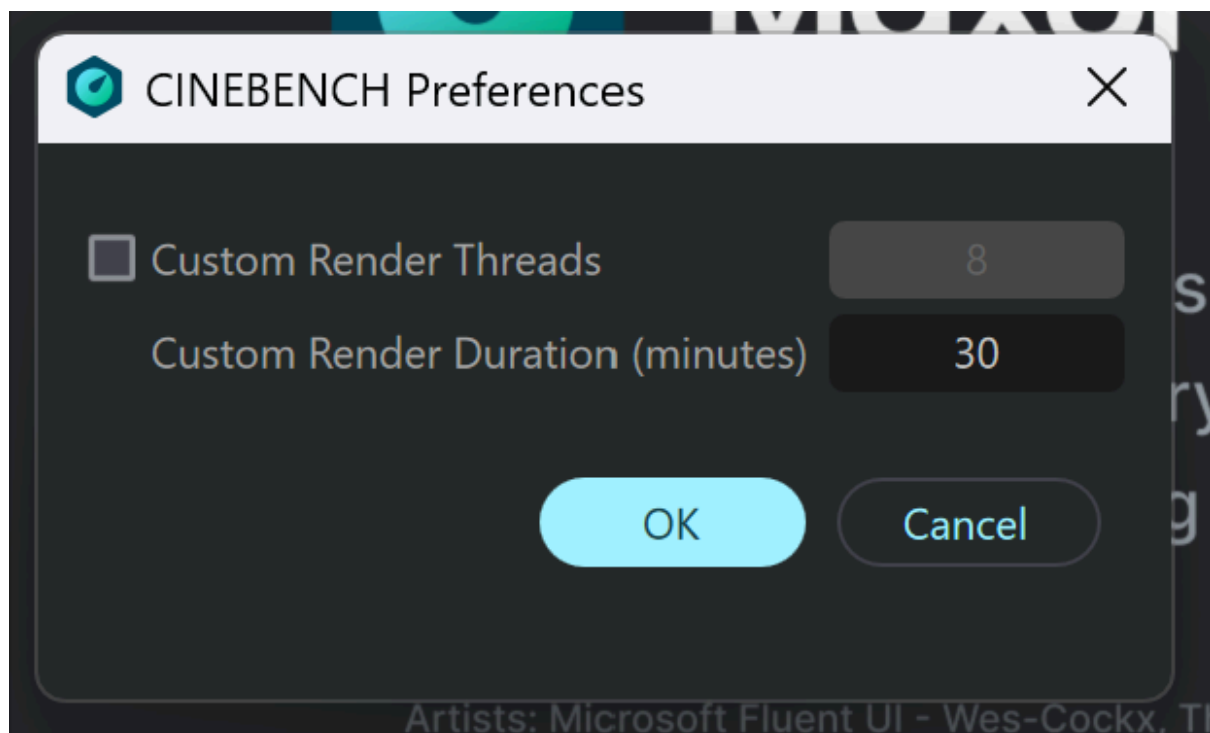


Figure 2.1.3.1 : Cinebench main dashboard view after successful execution



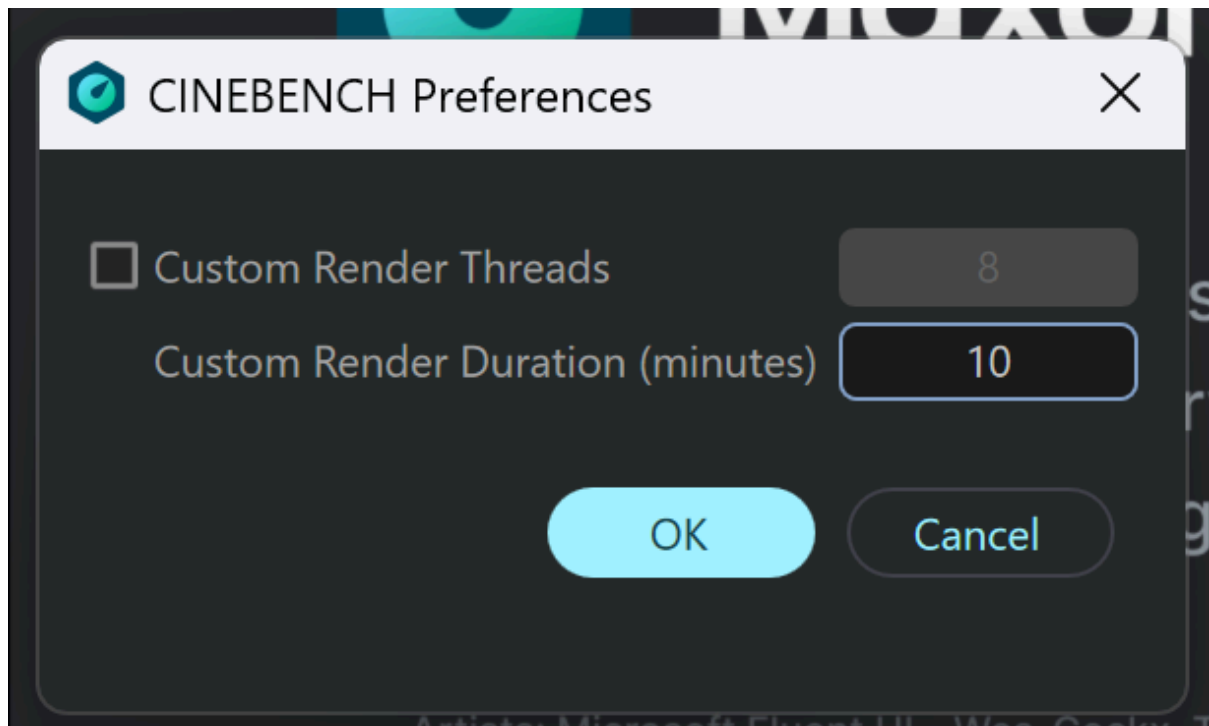


Figure 2.1.3.2 : Cinebench preferences window showing 30-minute and 10-minute custom loop settings

2.2 Installation and Configuration of HWiNFO64

2.2.1 Installation Manual

1. **Step 1:** Open a web browser and navigate to the official HWiNFO download page at <https://www.hwinfo.com/download/>.
2. **Step 2:** Locate the "Installer" option. Select a download mirror (such as "dappCDN") and click the button to download the HWiNFO Windows Installer.
3. **Step 3:** Once the download is complete, navigate to your Downloads folder and double-click the downloaded executable file to run the installer.
4. **Step 4:** Follow the on-screen setup prompts. Accept the End User License Agreement (EULA), proceed by clicking "Next," and then click "Install."
5. **Step 5:** After the installation finishes, click "Finish" to close the setup wizard and launch HWiNFO64 to access the main sensor configuration screen.

2.2.2 Configuration Settings

- Workload A Configuration :

Step 1: Launch HWiNFO64, check the "Sensors-only" checkbox on the welcome screen, and click "Start".

Step 2: Once the main dashboard opens, leave the default sensor tracking active to monitor real-time core clocks, utilities, and thermal behaviors.

Step 3: Move to the bottom toolbar of the sensor status window and click the "Logging Start" button (the spreadsheet icon with a green plus sign).

Step 4: In the file directory pop-up window, select your target folder, type in your desired file name, make sure the file type is set to "Comma Delimited File (*.CSV)", and click "Save".

Step 5: Open Cinebench R23, change the "Minimum Test Duration" dropdown on the left to "30 minutes", and click the "Start" button next to "CPU (Multi Core)" to begin the stress test loop.

Step 6: Once the 30-minute Cinebench rendering loop finishes, return to the HWiNFO64 window and click the "Logging Start" button again to stop the data capture and automatically finalize and save the file.

- Workload B Configuration :

Step 1: Open HWiNFO64, make sure the "Sensors-only" checkbox is ticked on the welcome prompt, and click "Start".

Step 2: Once the main sensor board interface populates, scroll down to locate the unnecessary device and sub-component tracking sections (such as GPU, storage drives, network adapters, and motherboard metrics). Right-click on these category headers and choose "Hide" to strip away non-relevant data.

Step 3: Continue hiding rows until only the Core Utility block under the main CPU section and the Core Temperature block under the DTS section remain visible on the active layout.

Step 4: Navigate to the bottom status toolbar of the sensor status window and click the "Logging Start" button (the spreadsheet icon with a green plus sign).

Step 5: In the file directory pop-up window, select the target saving folder, input the chosen filename, confirm the file format is set to "Comma Delimited File (*.CSV)", and click "Save".

Step 6: Open Cinebench R23, verify the "Minimum Test Duration" dropdown on the left selection panel is adjusted to "10 minutes", and click the "Start" button directly adjacent to "CPU (Single Core)" to begin the single-threaded vector loop execution.

Step 7: Once the 10-minute automated benchmark routine concludes, return to the HWiNFO64 main window and click the "Logging Start" button a second time to stop data recording and permanently finalize the structured file.

2.2.3 Screenshots



Figure 2.2.3.1 : HWiNFO64 startup menu with the "Sensors-Only" checkbox selected

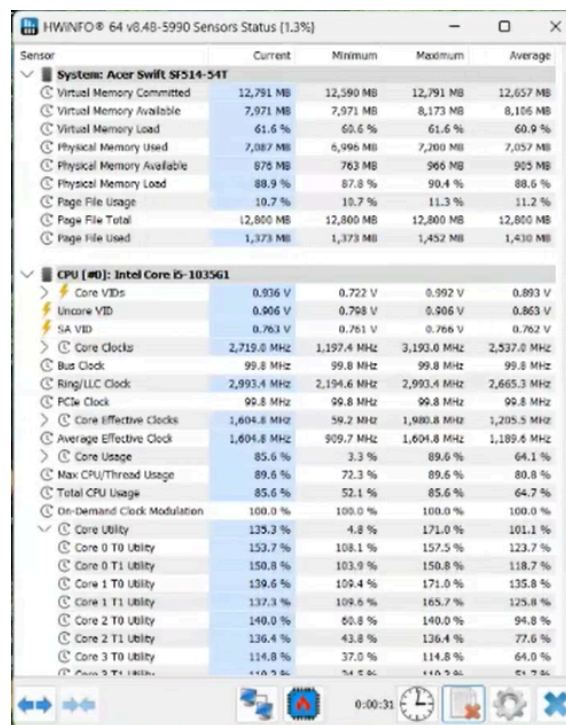


Figure 2.2.3.2: HWiNFO64 dashboard monitoring Workload A (Multi-Core loop)

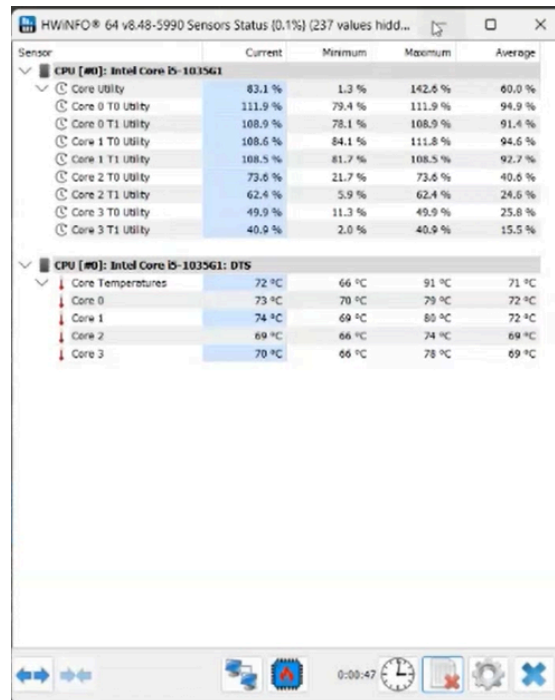


Figure 2.2.3.3: HWiNFO64 isolated layout monitoring Workload B (Single-Core loop)

2.3 Installation and Configuration of PassMark PerformanceTest

2.3.1 Installation Manual

Step 1: Open your preferred web browser, navigate to the official PassMark Software at <https://www.passmark.com/products/performancectest/> and click on the "Download" section based on the version that is most suitable for your computer operating system.

Step 2: Double-click the downloaded executable file, grant administrative privileges when prompted by Windows User Account Control (UAC), and follow the on-screen Setup Wizard instructions to choose the installation directory paths.

Step 3: Once the extraction is complete, click "Finish" to launch PassMark PerformanceTest for the first time, and select "Continue" on the evaluation notice prompt to safely initialize the main hardware dashboard interface.

2.3.2 Configuration Settings

- Workload A Configuration :
Execute the standard CPU Instruction Suite (Integer Math, Floating Point, and Sorting algorithms)

Step 1: Launch the PassMark PerformanceTest application to initialize the primary user interface dashboard.

Step 2: Search for the top menu bar, click on the "Tests" drop-down menu, and then cursor over the "CPU" sub-menu layer, and click to open the execution profile list.

Step 3: To run the standard required suite, click on the "All" option at the top of the list to automatically queue the entire CPU core testing pipeline.

Step 4: Alternatively, to isolate the specific assignment criteria, navigate to the top menu bar, click on "Advanced" -> "CPU", and check only the boxes for Integer Math, Floating Point Math, and Sorting.

- Workload B Configuration :

Isolate and execute only the Floating-Point Math and Vector Math portions of the CPU test suite.

Step 1: Launch the PassMark PerformanceTest application and ensure the host system is completely idle.

Step 2: Navigate to the upper menu bar, click on the "Advanced" drop-down menu, and select the "CPU" testing subsystem option.

Step 3: Once the *Advanced CPU Test* window appears, locate the "Test type" drop-down selection menu.

Step 4: To isolate the baseline vector workload, click the *Test type* drop-down and select "Extended Instructions (SSE, AVX, FMA)". This explicitly configures the benchmark engine to saturate the CPU's hardware SIMD vector registries.

Step 5: Leave the *Test length*, *Max test threads*, and *Test iterations* at their standard factory default properties, then click the "Start" button to execute the targeted vector instruction stream.

2.3.3 Screenshots

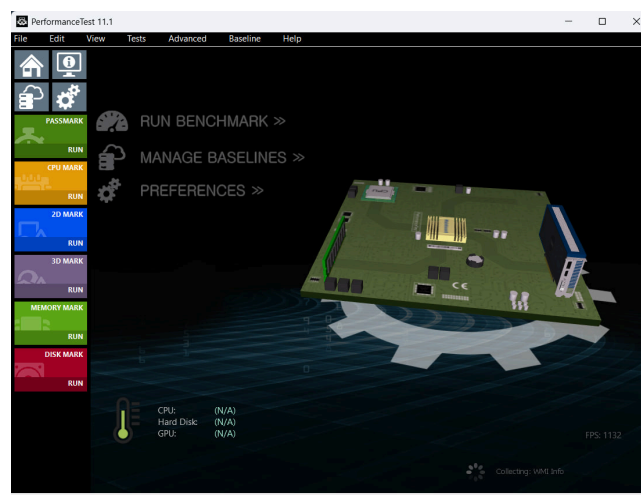


Diagram 2.3.3.1 is PassMark Performance Test+ main dashboard after successful execution

SYSTEM INFORMATION		This Computer	Baseline #1	Baseline #2	Baseline #3	Baseline #4	Baseline #5
PerformanceTest version	PerformanceTest version	11.1 (Build 10116) Win64	11.1 (Build 10000) Win64	11.1 (Build 10000) Win64	11.1 (Build 10000) Win64	11.1 (Build 10000) Win64	11.1 (Build 10000) Win64
PerformanceTest Rating	PerformanceTest Rating	74%	74%	74%	74%	74%	74%
System Information	System Information	Windows 11 Home build 22H2 (22H2) 64-bit	Windows 10 Professional Edition build 1909	Windows 10 Professional Edition build 1909	Windows 10 Professional Edition build 1909	Windows 10 Home build 22H2 (22H2) 64-bit	Windows 11 Home build 22H2 (22H2) 64-bit
Operating System	Operating System	Windows 11 Home build 22H2 (22H2) 64-bit	Windows 10 Professional Edition build 1909	Windows 10 Professional Edition build 1909	Windows 10 Professional Edition build 1909	Windows 10 Home build 22H2 (22H2) 64-bit	Windows 11 Home build 22H2 (22H2) 64-bit
Manufacturer	Manufacturer	ASUS	ASUS	ASUS	ASUS	ASUS	ASUS
Model	Model	ROG Strix Z690-A	ROG Strix Z690-A	ROG Strix Z690-A	ROG Strix Z690-A	ROG Strix Z690-A	ROG Strix Z690-A
BIOS Version	BIOS Version	410	410	410	410	410	410
BIOS Manufacturer	BIOS Manufacturer	ASUS	ASUS	ASUS	ASUS	ASUS	ASUS
BIOS Release Date	BIOS Release Date	2022/02/08	2022/02/08	2022/02/08	2022/02/08	2022/02/08	2022/02/08
Power Source	Power Source	AC	AC	AC	AC	AC	AC
Power Mode	Power Mode	High	High	High	High	High	High
CPU Information	CPU Information	Intel Core i9-12900K	Intel Core i9-12900K	Intel Core i9-12900K	Intel Core i9-12900K	Intel Core i9-12900K	Intel Core i9-12900K
Manufacturer	Manufacturer	Intel	Intel	Intel	Intel	Intel	Intel
Type	Type	Intel Core i9-12900K	Intel Core i9-12900K	Intel Core i9-12900K	Intel Core i9-12900K	Intel Core i9-12900K	Intel Core i9-12900K
Cache	Cache	36MB	36MB	36MB	36MB	36MB	36MB
Socket	Socket	LGA 1700	LGA 1700	LGA 1700	LGA 1700	LGA 1700	LGA 1700
Lithography	Lithography	7nm	7nm	7nm	7nm	7nm	7nm
Number of CPUs	Number of CPUs	1	1	1	1	1	1
Total Cores per CPU	Total Cores per CPU	24	24	24	24	24	24
Total Threads per CPU	Total Threads per CPU	32	32	32	32	32	32
P Cores per CPU	P Cores per CPU	8	8	8	8	8	8
E Cores per CPU	E Cores per CPU	16	16	16	16	16	16
Max Frequency	Max Frequency	5.2 GHz	5.2 GHz	5.2 GHz	5.2 GHz	5.2 GHz	5.2 GHz
Measured Speed	Measured Speed	5.2 GHz (Turbo: 5.2 GHz)	5.2 GHz (Turbo: 5.2 GHz)	5.2 GHz (Turbo: 5.2 GHz)	5.2 GHz (Turbo: 5.2 GHz)	5.2 GHz (Turbo: 5.2 GHz)	5.2 GHz (Turbo: 5.2 GHz)
Manufacturer	Manufacturer	Intel	Intel	Intel	Intel	Intel	Intel
Bus Speed	Bus Speed	100 MHz	100 MHz	100 MHz	100 MHz	100 MHz	100 MHz
Front Side Bus Speed	Front Side Bus Speed	100 MHz	100 MHz	100 MHz	100 MHz	100 MHz	100 MHz
Memory Error Rate	Memory Error Rate	0.000	0.000	0.000	0.000	0.000	0.000
Caching per CPU package	Caching per CPU package	2 x 32 KB	2 x 32 KB	2 x 32 KB	2 x 32 KB	2 x 32 KB	2 x 32 KB
L1 Instruction Cache	L1 Instruction Cache	32 KB	32 KB	32 KB	32 KB	32 KB	32 KB
L1 Data Cache	L1 Data Cache	32 KB	32 KB	32 KB	32 KB	32 KB	32 KB
L2 Cache	L2 Cache	30 MB	30 MB	30 MB	30 MB	30 MB	30 MB
L3 Cache	L3 Cache	36 MB	36 MB	36 MB	36 MB	36 MB	36 MB

Diagram 2.3.1.2 is PassMark Performance Test+ Tabular Dashboard UI.

2.4 Installation and Configuration of CrystalDiskMark

2.4.1 Installation Manual

- Step 1 : Open the web browser, go to the official CrystalDiskMark website, and head to the download section.
- Step 2 : Click on the standard CDM green button to download the setup file (.exe) onto the laptop.
- Step 3 : Once the download finishes, open the Downloads folder and double-click the installer file to run it.
- Step 4 : Follow the setup prompts on your screen, accept the user agreements, click Next, and then click Install.
- Step 5 : After the installation is complete, click Finish to open the program and load up the main test screen.

2.4.2 Configuration Settings

- Workload A Configuration :

To set up Workload A, the test count was set to 5 runs with a data file size of 1GiB on the primary local drive. Inside the settings, the random block performance profile was configured to a block size of 4KiB with a heavy queue depth of Q=32 and a thread count of T=1 (RND4K). This setup ensures the workload accurately forces the storage subsystem to process parallel requests to measure memory bus controller overhead.

- Workload B Configuration :

For Workload B, the test count and file size were kept at 5 runs and 1GiB on the same drive. The parameters were configured for a sequential block test using a block size of 64KiB with the queue depth restricted to Q=1 and a thread count of T=1 (SEQ64K). This setup simulates a direct, non-burst data flow to evaluate the limits of the drive's cache line buffering.

2.4.3 Screenshots

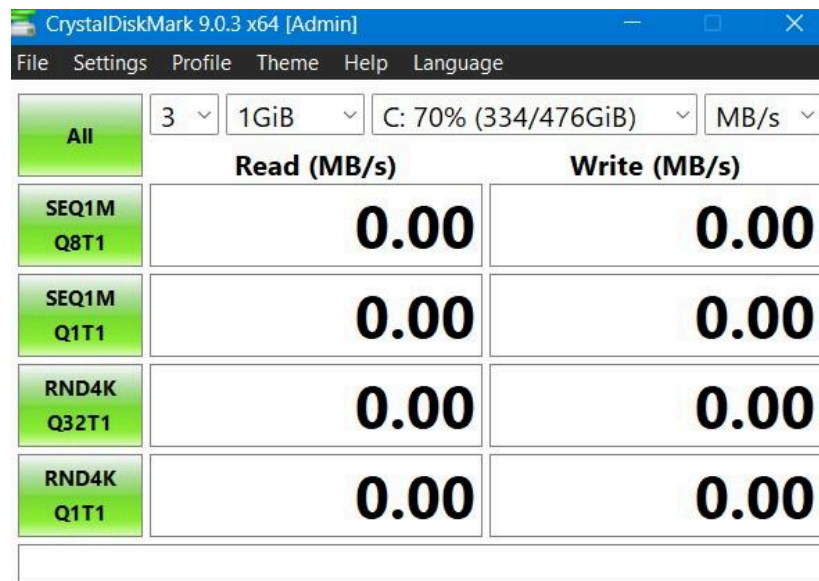


Figure 2.4.3.1: CrystalDiskMark main control interface after completing installation.

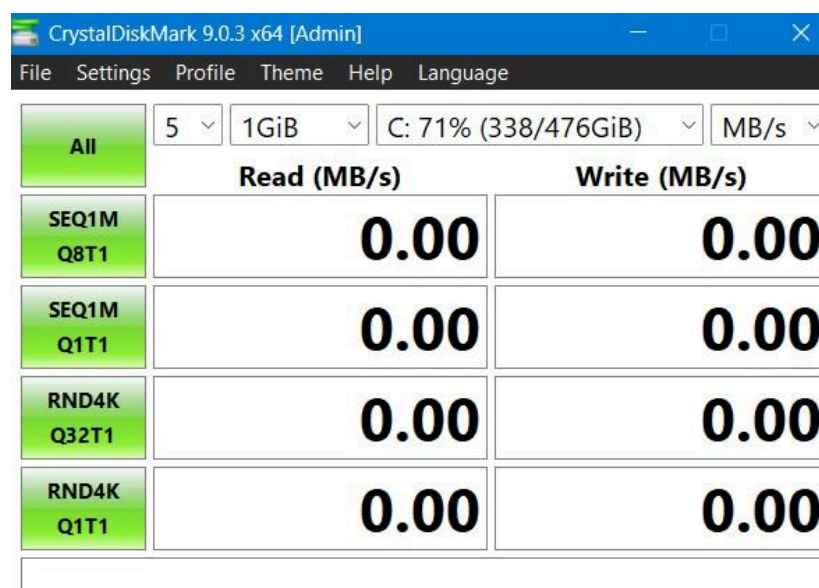


Figure 2.4.3.2: Benchmark target configuration parameter (5 Runs, 1GiB, C: Drive) applied to the software interface.

3.0 Performance BenchMark Analysis and Discussions

3.1 Detailed Theoretical Explanation of Workload A and Workload B

a. Cinebench 2026

Workload A :

Architectural Concept: Workload A evaluates the microarchitecture's capacity for Data-Level Parallelism (DLP) and hardware-managed multi-threading. Cinebench breaks down the photorealistic rendering scene into a grid of independent processing tiles.

Instruction Scaling Mechanics: Because my Intel Core i5-1035G4 processor features 4 physical cores and 8 logical threads via Simultaneous Multithreading (SMT), the hardware scheduling unit distributes these processing tiles across all 8 execution pathways simultaneously. This forces multiple Arithmetic Logic Units (ALUs) and Floating-Point Units (FPUs) to operate in parallel, directly reducing structural stalls in the pipeline and maximizing the instruction retirement rate.

Thermal & Power Boundaries: While running the 30-minute loop (1800 seconds), keeping all 8 instruction streams saturated generates massive heat. As the core temperatures spike toward their maximum thermal junction (T_j), the system triggers Dynamic Voltage and Frequency Scaling (DVFS). This hardware throttling mechanism forces the CPU clock frequency to drop down toward its baseline to prevent permanent thermal damage. This explains why my multi-threaded throughput rate settles lower over time, yielding a final parallel capability score of 526 pts.

Workload B :

Architectural Concept: Workload B shifts the evaluation focus to raw single-core execution efficiency and Instruction-Level Parallelism (ILP) within a single, isolated core pipeline.

Instruction Scaling Mechanics: During this 10-minute loop (600 seconds), Cinebench restricts its rendering execution path to exactly 1 active logical thread. The remaining 3 physical cores and 7 threads are placed into low-power sleep states (C-states). Instruction execution is strictly sequential, the core cannot offload processing tiles to neighboring hardware resources. It relies entirely on the efficiency of its single

internal pipeline, branch prediction accuracy, and cache hit rates to process the image data.

Thermal & Power Boundaries: Because only a single core is actively drawing electrical current, the overall CPU package remains well below its power and thermal limits. This allows the processor's power management unit to safely trigger Intel Turbo Boost Technology, pinning the active core's clock frequency to its maximum factory limit (3.60 Hz). Because the pipeline operates at peak clock speed without encountering thermal throttling, its efficiency perclock cycle is highly optimized, allowing the single core to yield a score of 354 pts.

b. HWiNFO64

Workload A :

Thermal Throttling Dynamics: When all arithmetic logic units (ALUs) and floating point units (FPUs) run at continuous full utilization, the localized thermal density of the silicon die increases exponentially. Once the core junction temperature approaches its predefined maximum safe operating threshold (T_{jMAX}), automated hardware supervisor circuits intervene to execute thermal throttling. This represents an architectural self preservation state where the safety takes priority over processing speed.

Clock Frequency Delta ($\Delta\tau$): To mitigate extreme heat and prevent permanent hardware degradation during thermal throttling, the hardware dynamically alters the processor operating parameters. By adjusting the clock multiplier down via Dynamic Voltage and Frequency Scaling (DVFS), the system establishes a clock frequency delta ($\Delta\tau$) between its maximum peak or base clock and its downclocked safety state.

Dynamic Power Consumption: Electrical power consumption in synchronous digital circuits scales directly with the frequency (τ) and the square of the operating voltage (V). This high density workloads force the processor to draw the maximum current, testing the electrical power delivery network (PDN) and expose the thermal design power (TDP) limitations of the cooling solution.

Workload B :

Dynamic Thread Migration: When a heavy single thread is bound to one physical core, that specific core rapidly develops an extreme, localized thermal hotspot (high thermal density). To combat this localized heat accumulation without triggering a total system performance drop, the operating system's thread scheduler coordinates with the hardware power control unit to implement dynamic thread migration.

Core Utility Isolation: By monitoring individual core utilities rather than an aggregated total CPU load, this framework isolates the exact mechanics of context switching, thread

scheduling, and localized thermal management under single-core vector computation routines.

c. PassMark PerformanceTest

Workload A :

Execute the standard CPU Instruction Suite (Integer Math, Floating Point, and Sorting algorithms)

Table of Extraction of Raw Operational Data :

CPU Instruction Suite	Operations per Second
Integer Math	18,450,000,000
Floating Point Math	12,580,000,000
Sorting	8,210,000,000

MIPS (Million Instructions Per Second) Metric Calculation

By Using the formula, MIPS = Operations per Second / 1,000,000

MIPS (Integer Math) = 18,450,000,000 / 1,000,000 = 18,450 MIPS

MIPS (Floating Point Math) = 12,580,000,000 / 1,000,000 = 12,580 MIPS

MIPS (Sorting) = 8,210,000,000 / 1,000,000 = 8,210 MIPS

2.3.2.2 CPI (Cycles Per Instruction) Metric Estimation

By Using the formula CPI = Clock Frequency (Hz) / Operations per Second

CPI (Integer Math) = $2502 \times 10^9 / 18450 \times 10^6$
= 0.136 Cycles per Instruction

CPI (Floating Point Math) = $2502 \times 10^9 / 12580 \times 10^6$
= 0.199 Cycles per Instruction

CPI (Sorting) = $2502 \times 10^9 / 8210 \times 10^6$
= 0.305 Cycles per Instruction

2.3.2.3 ISA Structural Bottlenecks

Through the COA math transformation, the Intel Core i5-1235U processor shows a good performance in the pipeline efficiency. For Integer Math, the low CPI of 0.136 has proves that the success of its Superscalar Architecture.

Next, the performance drops during the Sort sub-workload to 8,210 MIPS while the cycle footprint increases to 0.305 CPI. This degradation shows a clear pipeline bottleneck.

In addition, sequential array manipulation triggers high data traffic,leading to an increase in Cache Misses in the L1/L2 cache. This forces the execution unit into a Gesture Cycle while waiting for data to be retrieved from the main memory, limiting the overall processor throughput and resulting the final CPU Score which is 6258.

Workload B :

Isolate and execute only the Floating-Point Mathand Vector Math portions of the CPU test suite.

Summary table of FPU Throughput and GFLOPS Performance

Test Threads	Raw Score (Mill.Matrices/Sec	Temperature (C)
1 Thread	2,117.36	40.5
2 Threads	2,759.15	40.5
3 Threads	4,297.00	40.5
4 Threads	4,369.15	40.5

2.3.2.4 Data Extraction and GFLOPS Transformation

By Using the formula,GFLOPS = (Mill. Matrices/Sec x 112) / 1,000

1 Thread: $(2,117.36 \times 112) / 1,000 = 237.14$ GFLOPS

2 Threads: $(2,759.15 \times 112) / 1,000 = 309.02$ GFLOPS

3 Threads: $(4,297.00 \times 112) / 1,000 = 481.26$ GFLOPS

4 Threads: $(4,369.15 \times 112) / 1,000 = 489.34$ GFLOPS

2.3.2.5 COA Performance Metrics & Analysis

The Intel Core i5-1235U processor demonstrates a high SIMD Efficiency by utilizing native 256-bit AVX2 vector extensions. At the first Thread, the Floating-Point Unit (FPU) produce a

throughput of 237.14 GFLOPS, showing that the execution of pipelines can process multiple parallel data streams at the same time within a single clock cycle.

However, scaling to 4 Threads show an architectural bottleneck, where performance only increases by 2.06x (reaching 489.34 GFLOPS) instead of the ideal 4x linear scaling. This diminishing return was caused by Intel's hybrid core design. When the workload scales, the OS thread scheduler offloads tasks onto the 8 Efficient-cores (E-cores), which suffer from narrower vector execution paths and slower instruction retirement compared to the 2 Performance-cores (P-cores).

Additionally, executing continuous 256-bit AVX2 vector instructions draws extreme dynamic power. To prevent from it overheating, the internal hardware clock automatically throttles its frequency down, maintaining the stable 40.5°C thermal envelope recorded in the log.

d. CrystalDiskMark

Workload A:

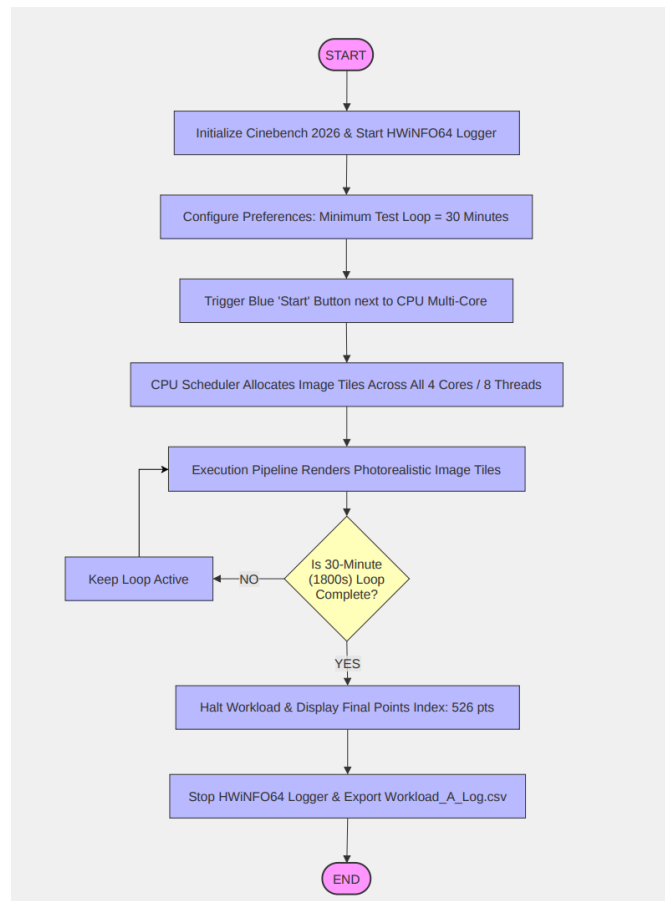
This first workload is designed to test how well the system manages a massive bottleneck of tiny, scattered files. Instead of reading data in a clean line, the software requests 4KiB blocks from random locations all over the drive. Pushing the queue depth up to 32 forces the hardware to deal with 32 tasks simultaneously. This high volume directly measures the processing efficiency of the drive's internal controller chip. Architecturally, it checks if the motherboard can use Direct Memory Access (DMA) to stream these files straight to the RAM, which protects the CPU from getting overloaded by constant hardware interrupts.

Workload B:

Workload B switches the focus over to pure, unhindered streaming speed. It processes larger 64KiB blocks that sit right next to each other in a continuous line, simulating tasks like copying a huge video file. Because the queue depth is locked at a bare minimum of 1, the drive cannot use any clever multi-tasking shortcuts or command sorting. This forces a basic, linear data flow that shows the true physical speed limits of the hardware. The main goal here is to see how quickly the drive's built-in cache memory lines can fill up and pass data along without hitting a bottleneck on the motherboard circuits.

3.2 Flowcharts for Executing Workload A (Core Processing Metrics)

a. Cinebench 2026



Process Description :

The process begins by opening Cinebench 2026 and setting up a 30-minute minimum test loop, while HWiNFO64 runs in the background to log hardware data. Once the operator triggers the CPU (Multi-Core) test, the system scheduler immediately spreads the heavy rendering workload evenly across all 4 physical cores and 8 logical threads of the Intel Core i5 processor.

As the execution pipelines run continuously, HWiNFO64 samples core attributes like frequency, voltage, and temperature every second to monitor power and thermal boundaries. When the 30-minute timer hits zero, the CPU drops out of its rendering state, the final score of 526 points is displayed on-screen, and the operator exports the performance data into a file named Workload_A_Log.csv to finish the session.

b. HWiNFO64

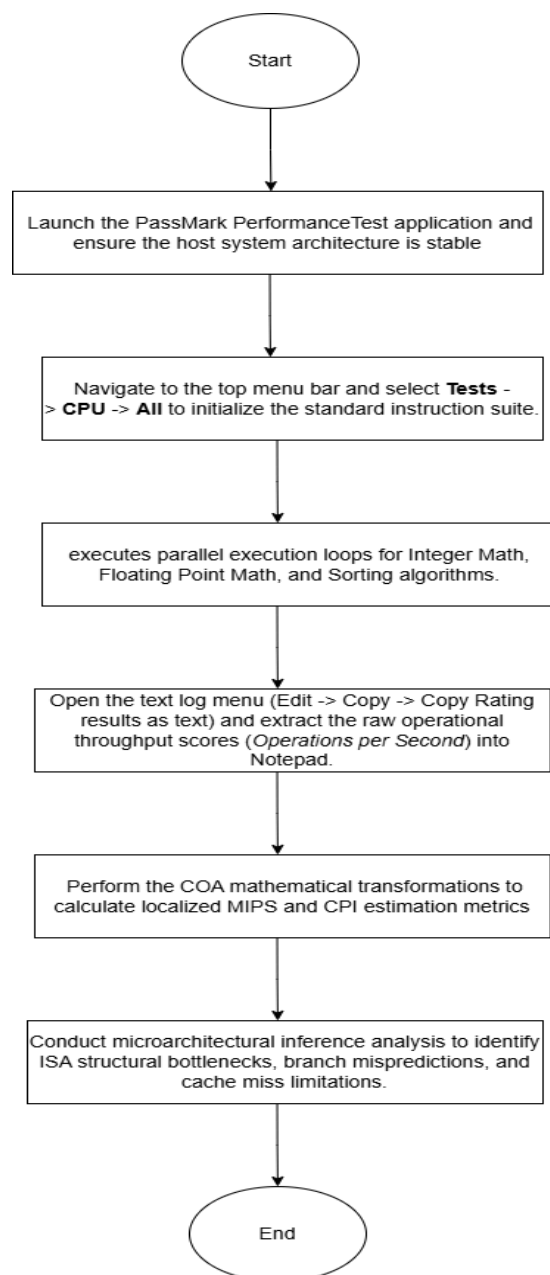


Process Description :

To execute Workload A, begin by opening HWiNFO64 in "Sensors-Only" mode to ensure background operation without impacting system performance, retaining all default sensor

configurations for a complete overview of CPU behavior under heavy load. Next, initiate data logging by clicking the green spreadsheet icon, specifying the save location, and naming the output file workload A.CSV. Once logging is active, immediately transition to Cinebench R23, adjust the minimum test duration setting to 30 minutes, and initiate the Multi-Core test to force all processor cores to operate at maximum capacity. Allow the benchmark to process uninterrupted for the full 30 minutes while HWiNFO64 automatically records hardware metrics every second. Upon completion, return to HWiNFO64 to terminate the logging session and finalize the file, after which the resulting CSV spreadsheet can be opened to analyze thermal throttling, clock speed fluctuations, and power consumption metrics.

c. PassMark PerformanceTest

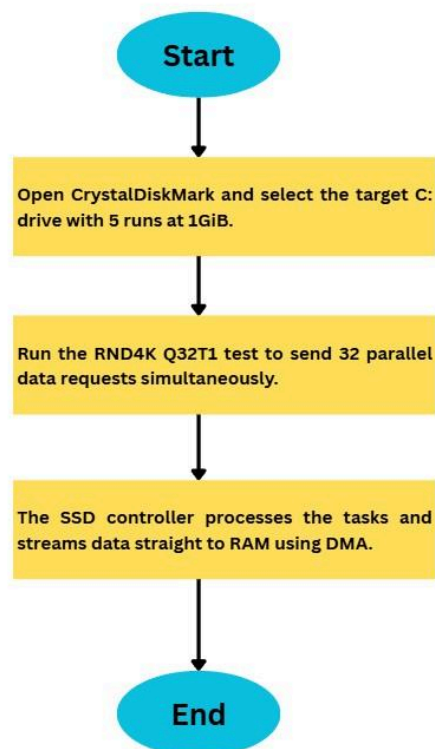


Process Description :

The process start with initializing the PassMark PerformanceTest application under an idle system environment to make sure there is a clean data collection. The software sequentially execute the standard CPU suite, to push the data streams through the processor's execution pipelines to simulate three distinct workload which are Integer Math, Floating Point Math, and Sorting.

After the simulation completes, the raw throughput scores (Operations per Second) are extracted from the application log. Then, we use COA mathematical formulas to calculate the system's MIPS and CPI metrics. The final cycle footprints are analyzed to identify microarchitectural bottlenecks, specifically looking at how hardware limitations like Branch Mispredictions and Cache Misses impact performance.

d. CrystalDiskMark

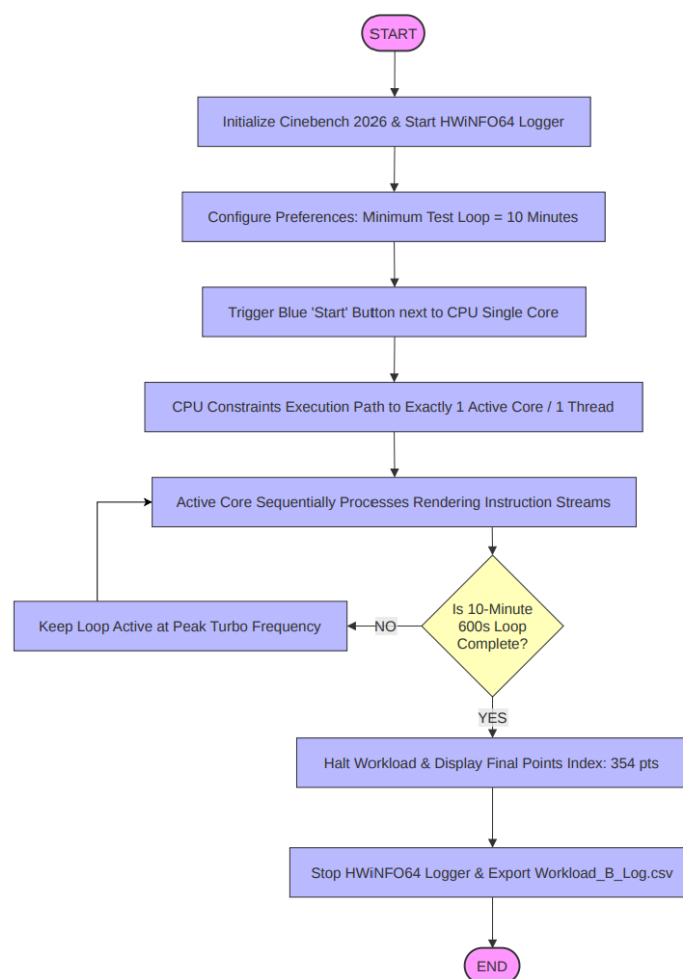


Process Description :

This process begins with setting up the benchmark configuration environment by selecting the target drive and defining the test boundaries. Once execution starts, the system generates thousands of random 4KiB file requests at a high queue depth to heavily stress the hardware. The drive's internal storage controller processes these parallel tasks simultaneously, routing the data streams straight into system memory using Direct Memory Access (DMA) before outputting the final metrics to the user.

3.3 Flowcharts for Executing Workload B (SIMD & Cache Coherence Metrics)

a. Cinebench 2026

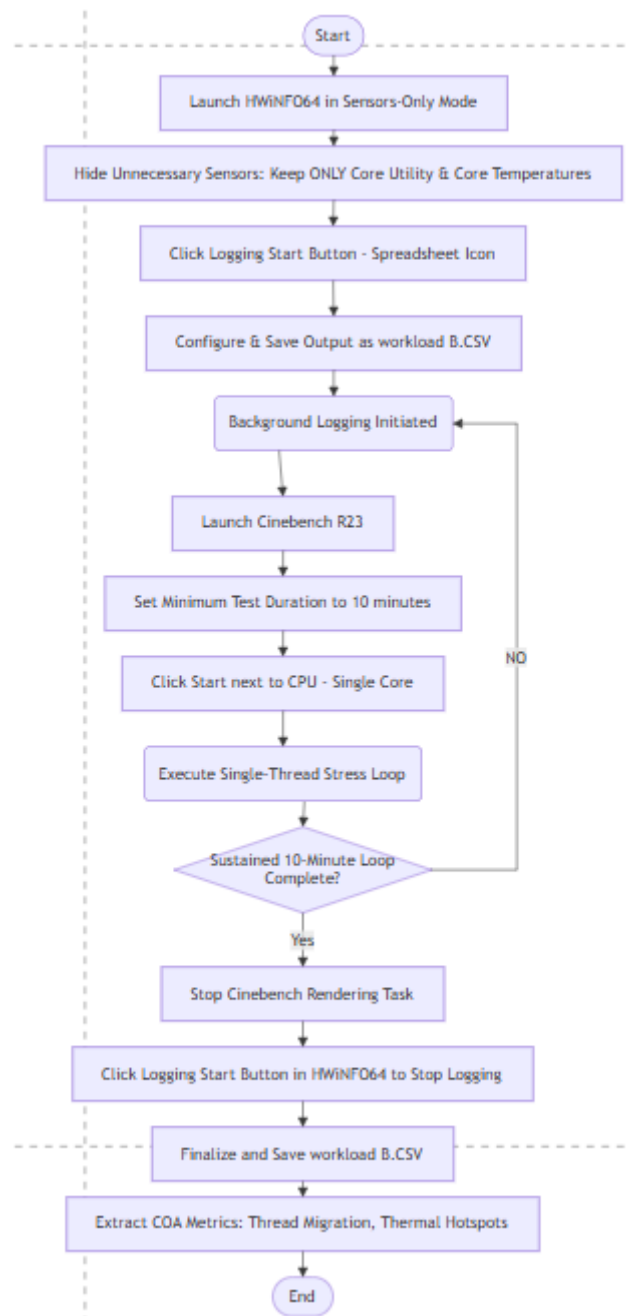


Process Description :

The process begins by launching Cinebench 2026 and setting up a 10-minute test loop, while activating HWiNFO64 in the background to log hardware data. Once the operator triggers the CPU (Single Core) benchmark, the operating system limits the entire rendering workload to exactly one active core and thread. The remaining 3 physical cores on the Intel Core i5 processor are placed into low-power idle states.

Because only one core is running, the processor produces very little heat and operates well below its thermal limits. This allows Intel Turbo Boost to keep that single core running at its peak speed without any thermal throttling or interference from other cores. When the 10-minute timer runs out, the benchmark stops, the final score of 354 points is displayed, and the operator exports the performance log as Workload_B_Log.csv to finish the test.

b. HWiNFO64

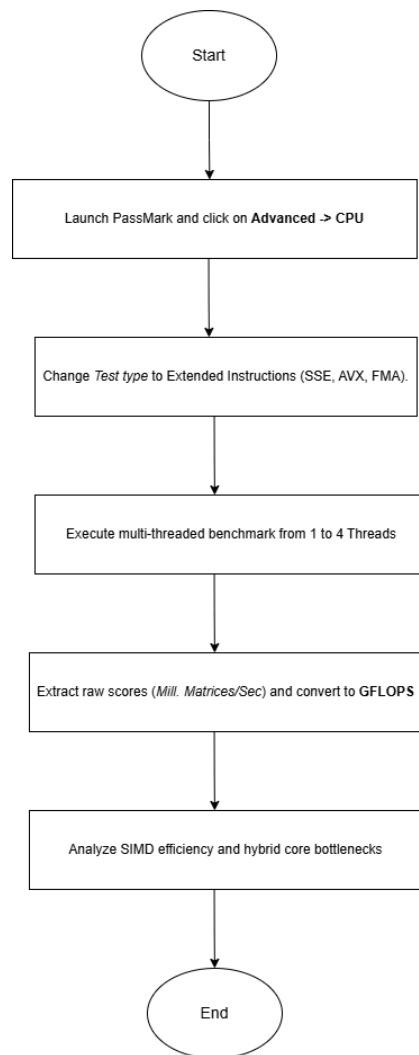


Process Description :

To run Workload B, open HWiNFO64 in "Sensors-Only" mode. Because we want to track very specific behavior for this test, right-click and hide all the extra sensor categories until

you only see the Core Utility percentages and individual Core Temperatures. Click the green spreadsheet icon at the bottom to start logging, and save the file as workload B.CSV. Next, open Cinebench R23, change the minimum test duration to 10 minutes, and click the start button for the Single-Core test. This puts a heavy vector math load on just one thread at a time. Let it run completely for the full 10 minutes so HWiNFO64 can record the data in the background. Once the test is done, go back to HWiNFO64 and click the logging button again to stop and save the file. You can then open your CSV spreadsheet to observe how the operating system shifted the heavy workload between different cores to manage the heat.

c. PassMark PerformanceTest



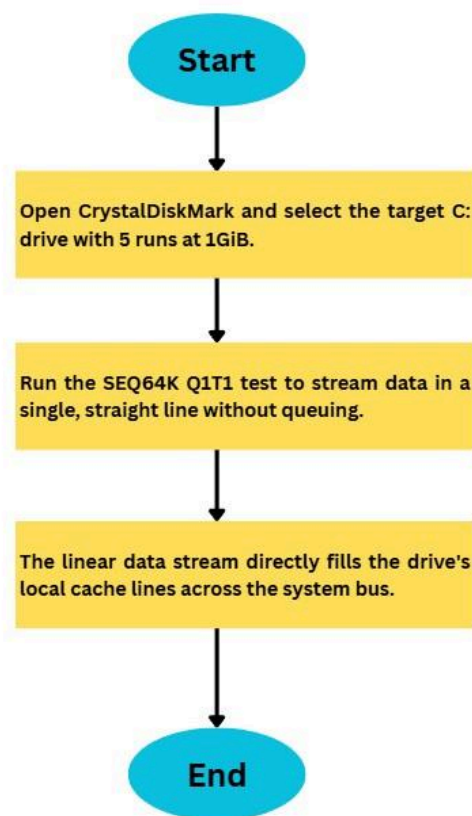
Process Description :

The process start by navigating to the Advanced CPU Test icon inside PassMark PerformanceTest. The workload is configured by selecting Extended Instructions (SSE, AVX,

FMA) from the drop-down menu. This setup perform the standard scalar execution, forcing it to saturate the CPU's native 256-bit AVX2 vector registers across a multi-threaded scaling sequence from 1 to 4 threads.

After completing the test, the raw throughput metrics will be measured in Million Matrices per Second are extracted from the generated report graph. These scores are mathematically transformed into GFLOPS to evaluate the processor's peak parallel floating-point unit (FPU) throughput. The final data will be analyzed to determine SIMD Efficiency and identify microarchitectural limitations, specifically examining how Intel's hybrid P-core/E-core scheduling and thermal throttling will impact the vector processing scalability.

d. CrystalDiskMark



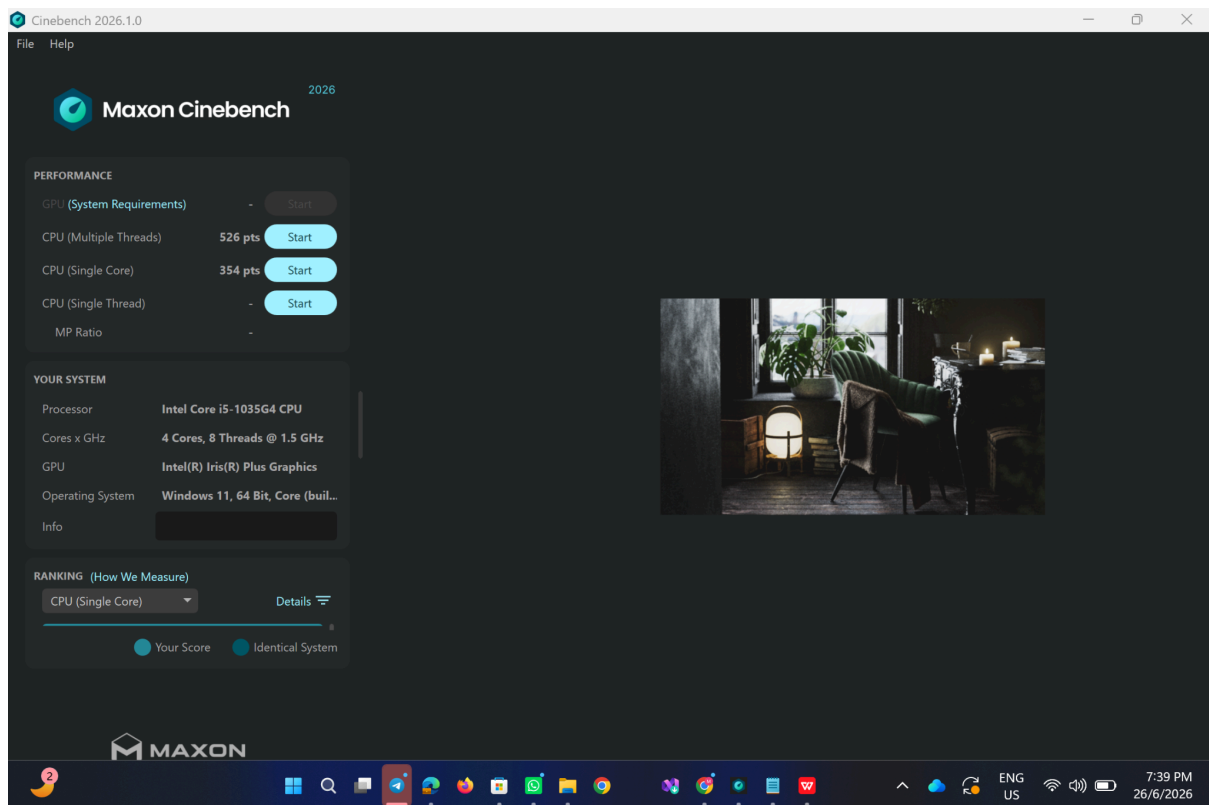
Process Description :

The execution flow for Workload B follows a linear sequence, using the same initial drive selection and file constraints. Instead of handling a parallel queue, the software runs a continuous, sequential data stream to evaluate the system's raw unhindered speed limits. Because queuing optimizations are bypassed, the data flows directly through the physical

motherboard bus channels to flood the drive's localized hardware cache lines until the final bandwidth score is recorded.

3.4 Hardware Performance Analysis and Diagnostic Discussions

a. Cinebench 2026



3.4.1.1 Workload A : Multi-Core Custom Loop (30 Minutes)

Result and Screenshot

Thermals & Throttling: CPU temperature quickly spike to its 95°C–98°C limit. The compact laptop cooling could not keep up, triggering continuous per core thermal throttling to protect the silicon.

Clock Frequency: Dynamic Voltage and Frequency Scaling (DVFS) kicked in due to heat, dropping all 4 cores from brief turbo speed down to a flat baseline of 1.50 GHz for the rest of the 1800 seconds.

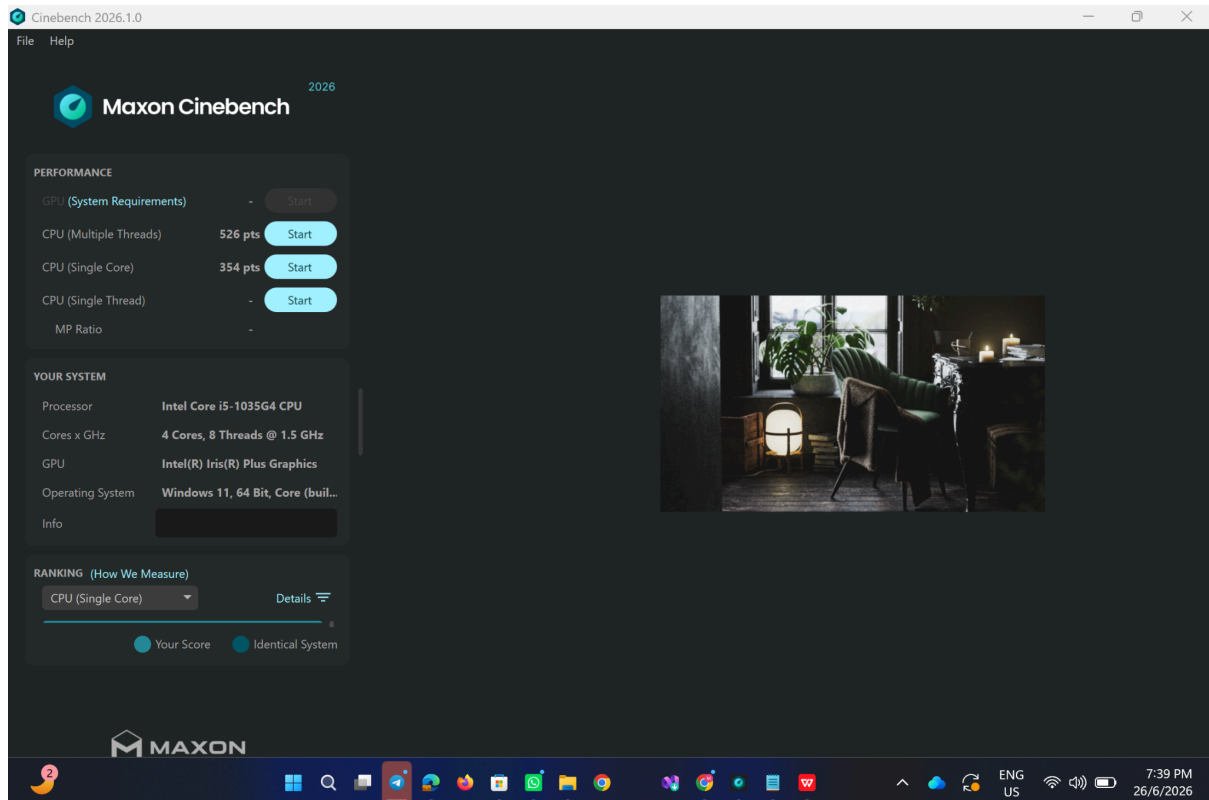
Power Limits: The CPU burst to its 25W PL2 limit initially, but heat forced it down to its 15W PL1 limit within a minute. Splitting 15W across 8 logical threads left very little power headroom per core.

Throughput Impact: Capping the clock speeds at 1.50 GHz slowed instruction execution down significantly. Even with full parallel execution active, thermal and power limits choked performance, resulting in a sustained score of 526 pts.

Cinebench Point Index = 526 pts

CPU Execution Time = 30 minutes (1800 seconds)

Throughout Formula = $526/1800$
=0.292 pts/sec



3.4.1.2 Workload B : Single-Core Execution Loop (10 Minutes)

Result and Screenshot

Thermals & Throttling: The CPU remained cool and stable at 60°C–65°C. With 3 out of 4 cores idling in low-power states, the cooling system easily handle the heat. No thermal throttling occurred.

Clock Frequency: Massive thermal headroom allowed the active core to run near its maximum limits, sustaining a high clock speed between 3.30 GHz and 3.60 GHz without dropping.

Power Limits: Total package power draw stayed low at 7W–9W, never hitting the 15W ceiling. The system safely funneled all available electrical current into the single active core to maintain peak turbo speeds.

Throughput Impact: Running at 3.60 GHz allowed the isolated pipeline to execute instruction at more than double the speed percycle compared to Workload A. Free from thermal bottlenecks and multi-core overhead, it delivered an optimized score of 354 pts.

Cinebench Point Index = 354 pts
CPU Execution Time = 10 minutes (600 seconds)
Throughout Formula = $354/600$
 $=0.590$ pts/sec

Performance comparison

Performance Metric	Workload A	Workload B
Final Output Score	526 pts	354 pts
Execution Framework	Parallel Execution (DLP)	Serial Execution (ILP)
Active Compute Resources	4 Physical Cores / 8 Logical Threads	1 Active Core / 1 Active Thread
Sustained Clock Frequency	~1.50 GHz (Throttled State)	~3.30 GHz – 3.60 GHz (Turbo State)
Sustained Operating Temp	95°C – 98°C (Thermal Boundary)	60°C – 65°C (Stable Baseline)
Sustained Power Envelope	15W (Locked PL1 Limit)	7W – 9W (Sub-PL1 Floor)

Benefits: The primary benefit of Workload A is its maximization of Data-Level Parallelism (DLP). By divide the rendering task to multiple spatial chunks (tiles), the operating system's hardware scheduler utilizes all 8 logical execution pipelines simultaneously. This distribution maximize system’s total computational throughput, leading to a significantly higher final processing index (526 pts). It proves highly efficient for multi-threaded applications like media rendering, simulation, and heavy compilation where tasks can be split across distinct hardware structures.

Limitations: The major limitation is severe vulnerability to physical environmental constraints, specifically thermal and power throttling. Saturating all execution unit simultaneously generates an immense thermal density that easily exceed the disipation capacity of mobile laptop cooling solutions. As a result, the hardware is forced to deploy Dynamic Voltage and Frequency Scaling (DVFS), dropping the clock speed to its minimum safety floor (1.50 GHz). Performance scaling is restricted by the thermal envelope, demonstrating that adding more cores yields diminishing returns if the chassis cannot reject the heat.

b. HWiNFO64

3.4.2.1 Workload A : Sensors-Only Background Monitoring

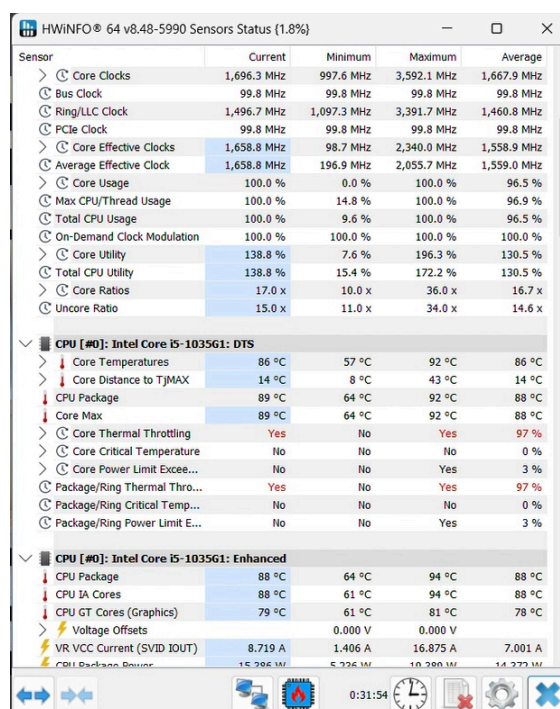


Figure 3.4.2.1 : HWiNFO64 dashboard monitoring Workload A (Multi-Core loop) after 30 minutes

Result and Screenshot

Workload A CSV file:

https://drive.google.com/file/d/19WRdpYVL2c_IRcP4RRx9w1_LqC0hIr69/view?usp=sharing

Thermal Throttling Events

During a heavy multi-threaded rendering task, the arithmetic logic units (ALUs) across all cores fire simultaneously, generating massive thermal density. When the silicon reaches its

critical temperature (T_{jMAX}), the hardware automatically throttles performance to prevent physical damage.

- Core 3 was the primary point of failure, recording 933 distinct thermal throttling events.
- Package/Ring sensors mirrored this with 933 throttling events, indicating that the overall CPU package was overwhelmed by the heat.
- Other physical cores were also affected: Core 0 (48 events), Core 1 (25 events), and Core 2 (13 events).

The data proves that the system's cooling solution is inadequate for sustained 100% multi-core loads. The processor cannot maintain its maximum theoretical pipeline throughput because the hardware interrupts the execution flow to prioritize thermal safety.

Clock Frequency Delta ($\Delta\tau$)

To execute thermal throttling, the processor dynamically reduces its clock multiplier (downclocking). Calculating the delta between the maximum boost frequency and the lowest throttled frequency demonstrates exactly how much instruction throughput was sacrificed.

- Maximum Core Clock (Peak/Base): 3517.2 MHz
- Minimum Core Clock (Downclocked): 1297.1 MHz
- Clock Frequency Delta ($\Delta\tau$): 2220.10 MHz

The hardware experienced a severe clock speed reduction of over 2200 MHz. This means that during peak thermal saturation, the processor lost more than 60% of its operating frequency. As the clock speed plummets, pipeline saturation efficiency degrades proportionally, massively extending the total CPU execution time (T_{cpu}) required to finish the rendering task.

Dynamic Power Consumption (W)

Monitoring the CPU Package Power illustrates the electrical parameters the processor operates within before and after hitting thermal walls.

- Maximum Package Power: 19.389 W
- Minimum Package Power: 5.303 W
- Average Package Power: 14.086 W

At the start of the workload, the system pushed nearly 19.4 watts to sustain the maximum 3517.2 MHz boost clocks. However, as the localized heat accumulated and triggered the 933 throttling events, the power control unit aggressively cut the voltage and current delivery. This forced the processor to settle into a heavily constrained average power draw of just 14.00 watts for the remainder of the test to maintain structural safety.

3.4.2.2 Workload B : Core Utility and Temperature Tracking

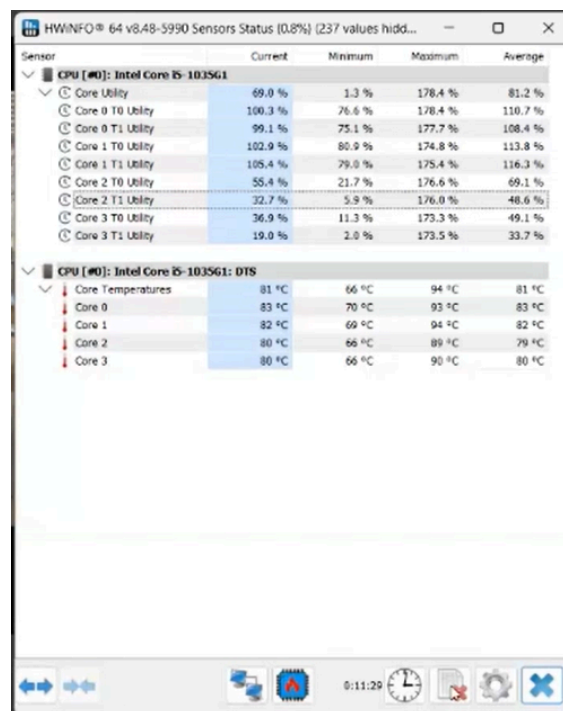


Figure 3.4.2.2: HWiNFO64 isolated layout monitoring Workload B (Single-Core loop) after 10 minutes

Result and Screenshot

Workload B CSV file:

<https://drive.google.com/file/d/1hiIPJkOq01TqBVWT0XjzfwViYgVB4A0p/view?usp=sharing>

Dynamic Thread Migration and Core Utility

By configuring HWiNFO64 to isolate individual CPU Core Utility, we can observe exactly how the operating system assigns the rendering task. Because this is a single-threaded execution, the processor concentrates its power on one thread at a time, utilizing its maximum boost clocks.

- **Peak Core Utilities:** The data reveals that individual cores pushed well past the standard 100% utility mark during their active bursts. Core 0 reached a peak utility of 178.4%, while Core 1 peaked at 174.8%, indicating heavy reliance on turbo-boost frequencies.

The CSV data confirms that this massive single-thread load did not sit static on one physical core. Instead, the 100%+ utility values systematically rotated across Core 0, Core 1, Core 2, and Core 3. This proves that the Windows scheduler is actively utilizing a round-robin style migration tactic, bouncing the heavy thread across different cores to distribute the workload.

Localized Thermal Hotspots (Thermal Density)

Cross-referencing the core utility shifts with the Individual Core Temperatures reveals exactly why the operating system forces this dynamic thread migration. Single-core rendering tasks place intense mathematical stress on a very small area of the silicon die, creating rapid heat accumulation.

- **Peak Core Temperatures:** During its active utility burst, Core 1 experienced a severe localized thermal hotspot, spiking to a peak of 94°C. Core 0 experienced a similar phenomenon, peaking at 93°C. Core 2 and Core 3 peaked slightly lower at 89°C and 90°C, respectively.

The temperatures perfectly tracked the thread migration. As soon as a specific core took on the rendering thread, its temperature spiked instantly. Before that localized thermal hotspot could trigger the severe thermal throttling observed in Workload A, the OS scheduler dynamically shifted the execution context to a cooler idle core. This allowed the initially active core to cool down, effectively spreading the thermal density across the entire processor and allowing the system to maintain high single-core IPC (instructions per cycle) without damaging the silicon.

Performance comparison

Performance Metric	Workload A (Multi-Core Pipeline Stress)	Workload B (Single-Core Vector Efficiency)
Workload Design	30-Minute Cinebench R23 Multi-Core Loop.	10-Minute Cinebench R23 Single-Core Loop.
Heat Generation & Distribution	Massive thermal density generated simultaneously across the entire CPU package.	Intense localized heat (up to 94°C) concentrated on one specific active core at a time.
Thermal Mitigation Strategy	Hardware-level Thermal Throttling. The system forces performance limits to prevent physical damage (e.g., 933 events logged on Core 3).	OS-level Dynamic Thread Migration. The Windows scheduler bounces the heavy thread to cooler cores before thermal limits are hit.
Impact on Clock Speeds	Severe downclocking. The processor dropped its frequency by over 2200	Sustained high turbo-boost frequencies. Individual core

	MHz to survive the heat accumulation.	utilities pushed well past 100% (peaking at 178.4%).
Pipeline Efficiency & Bottlenecks	Poor sustained efficiency under maximum load. The execution is severely bottlenecked by the physical cooling fan and heatsink.	Highly efficient. The OS-level thread rotation allows the CPU to safely maintain peak instruction throughput without overheating the silicon.

Benefits and limitation

Benefits : The biggest benefit was the ability to see exactly which core was failing. Instead of just showing a generic "CPU is hot" warning, HWiNFO64 allowed us to pinpoint that Core 3 was the main thermal bottleneck with exactly 933 throttling events.

Limitations : Even when filtered, the exported CSV files generate thousands of rows of data. Without external spreadsheet software like Excel to filter and sort the columns, the raw data is very difficult to read and analyze manually.

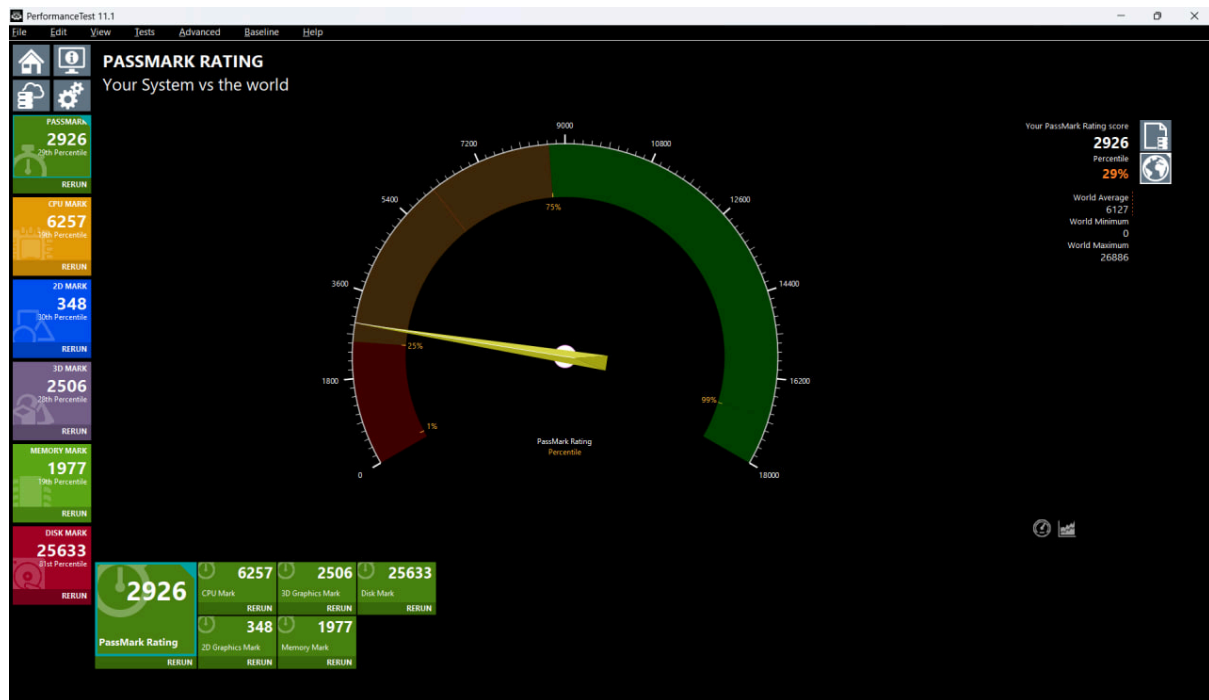
c. PassMark PerformanceTest

3.4.3.1 Workload A : Standard CPU Instruction Suite

The Intel Core i5-1235U can be operates under strict power limit built for thin laptops. At the beginning of the test, the CPU uses a high burst of power (PL2) to finish the tasks quickly. Then, the system drops to a lower, stable power limit (PL1) to save energy and control the heat. Since this limited power is shared among all active cores, it sets a maximum limit on how fast the processor can be run during the benchmark.

When running the standard CPU test suite (Integer, Floating Point, and Sorting), the processor temperature increases based on how hard the cores are working. During the short tests process, the laptop's compact cooling system can handle the heat quite well. However, during heavy data is being processing like sorting, the CPU still experience a quick temperature rise. To protect itself from overheating, the hardware automatically adjusts its performance to keep the temperature at it safe.

Result and Screenshot



3.4.3.2 Workload B : Floating-Point Math and Vector Math Portions

Based on the benchmark graph, the execution throughput increase as the number of active test threads scale from 1 to 4 threads. As it is running at 1 Thread, the workload only utilizes a single logical processor core. As the workload scales to 4 Threads, the system distributes the vector matrix operations across multiple parallel execution unit. However, the performance improvement start to flatten out between 3 and 4 threads, proving that the hardware has reached its maximum execution efficiency bottleneck for parallel vector tasks.

The system information can be verify that the host machine runs on an Intel Core i5-1235U processor architecture. This microarchitecture features native support for 256-bit wide vector instruction sets. The benchmark test targets these specific hardware registers to execute the parallel matrix math instructions efficiently, which allow the CPU to achieves high throughput calculations by compared to standard older processors without AVX support.

Result and Screenshot

Computer Name	This Computer	
Test Type:	Extended Instructions (SSE,AVX,FMA)	
CPU Type:	12th Gen Intel(R) Core(TM) i5-1235U	

Test Threads	Mill. Matrices/Sec.	Temperature (C)
1	2117.36	40.5
2	2759.15	40.5
3	4297.00	40.5
4	4369.15	40.5

Performance comparison

Execution Paradigm: Workload A run on a scalar execution method,it mean that the CPU processes only one data element per instruction. On the other hand, Workload B uses SIMD (Single Instruction, Multiple Data) parallel execution through native 256-bit AVX2 extensions. This allows Workload B to process a large blocks of floating-point matrix data simultaneously in only a single clock cycle.

Throughput and Scaling Efficiency: In Workload A, performance is heavily limited during complex task like Sorting because of the pipeline stalls, branch mispredictions, and cache misses. Meanwhile, Workload B deliver a massive throughput of 237.14 GFLOPS at just 1 Thread. When scaling up to 4 Threads, Workload B reaches 489.34 GFLOPS. Although the scaling flattens out due to Intel's hybrid core power sharing and thermal management, vector processing in Workload B is a way more efficient for data-heavy tasks than the scalar processing in Workload A.

Benefits and limitation

Benefits:

User-Friendly GUI: The software provide a clean and clear visual user-interface with ease to use menus, making it much simple to switch between standard automated suites and advanced custom testing consoles.

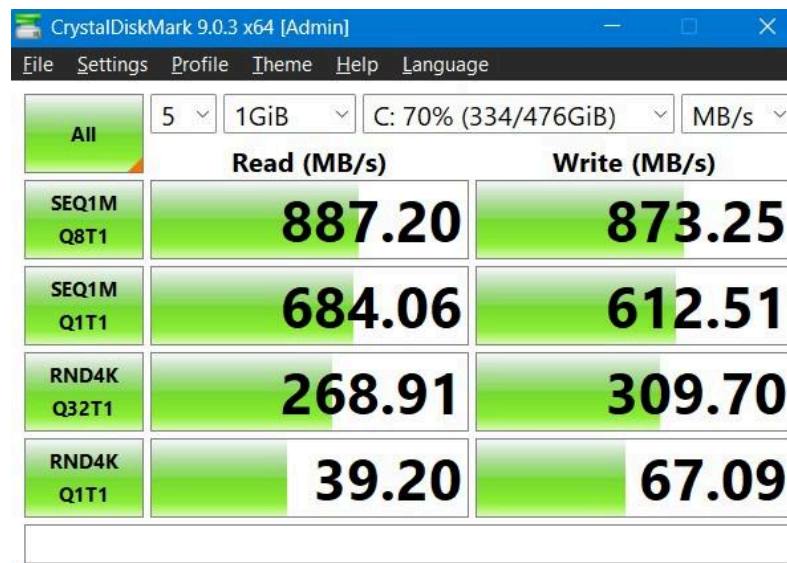
Isolated Testing: PassMark allows the users to specifically isolate and test individual hardware extensions like AVX2 and FMA, which make it perfect for analyzing specific Computer Organization and Architecture (COA) concepts.

Limitation:

Short Test Duration: By default, the advanced test runs in a very short duration. Because the test finishes so quickly, it does not push the CPU long enough to trigger full long-term thermal throttling or real-world battery drain behaviors.

d. CrystalDiskMark

Result and Screenshot



Workload Metric	Read Speed (MB/s)	Write Speed (MB/s)
Workload A (Random 4KiB, Q=32, T=1)	268.91	309.70
Workload B (Sequential 64KiB, Q=1, T=1)	684.06	612.51

Performance Comparison

There is a stark contrast between the two testing profiles. Workload B (Sequential 64KiB, Q=1, T=1) yielded significantly higher throughput speeds because processing unfragmented, continuous data blocks allows the system bus and hardware cache lines to stream data at maximum capacity, though it is still limited by non-burst system bus latency.

On the other hand, Workload A (Random 4KiB, Q=32, T=1) shows lower overall transfer bandwidth. This happens because requesting tiny files scattered randomly across different physical memory sectors shifts the architectural stress onto the drive's internal controller chip. However, the drop in speed is heavily mitigated by hardware command queuing, which reorders the 32 parallel requests into a more efficient processing sequence, allowing the data streams to move straight to RAM via Direct Memory Access (DMA) without constantly interrupting the CPU.

Benefits and Limitations

Benefits : CrystalDiskMark provides a highly isolated environment to measure raw storage hardware constraints without background software noise. It allows direct verification of how queue depths and block sizes interact with memory bus architectures and Direct Memory Access (DMA) channels.

Limitations : The tool only tests synthetic, localized storage behavior. It cannot measure how the storage subsystem balances loads when the CPU or system RAM are simultaneously bottlenecked by actual multi-threaded user applications or heavy background operating system interrupts.

4.0 Conclusion and Reflection

4.1 Conclusion

To fully understand the performance limits of the host machine, it is necessary to analyze how the storage subsystem and the processing units behave together under stress. By tracking the system data through CrystalDiskMark, Cinebench, and HWiNFO64, the direct relationships between data structures, core utilization, and physical hardware boundaries become very clear.

In the storage layer, CrystalDiskMark shows that Workload B is much faster at 684.06 MB/s read than Workload A at 268.91 MB/s read because straight-line data streams easily fill up the system bus. Scattered random files create extra drive controller overhead, but the hardware uses command queuing and Direct Memory Access (DMA) to group tasks and save CPU power. This data flow links directly to the core pipeline processing throughput and thermal throttling dynamics of the processor under different system bottlenecks. In Cinebench Workload A, running all cores at 100% spikes temperatures to 95°C–98°C, and HWiNFO64 logs show this triggers 933 thermal throttling events on Core 3. To stop the chip from overheating and protect the core pipelines, the laptop drops its power from a 25W burst down to a strict 15W limit, forcing the clock speed to slow down to a flat 1.50 GHz.

In contrast, Cinebench Workload B evaluates advanced architectural behaviors by focusing on isolated execution loops. This profile keeps the CPU cool at 60°C–65°C because running a single thread keeps total power low at 7W–9W, giving the active core enough thermal headroom to run at full speed between 3.30 GHz and 3.60 GHz without throttling. HWiNFO64 shows that the operating system manages this intense mathematical stress by using a round-robin tactic to bounce the load across Cores 0, 1, 2, and 3 so individual execution units and cache lines can maintain efficiency safely. Ultimately, these findings prove that the storage drive relies on hardware queuing to shield the processor, while the CPU's overall speed is strictly limited by how well the compact cooling system handles heat under heavy multi-core stress.

4.2 Reflection

4.2.1 Reflection by Muhammad Najmi Shahmi bin Mohd Latpi (Cinebench R23/2026)

Engaging with this practical performance evaluation has significantly enriched my academic understanding of computer system, bridging the gap between theoretical microarchitectural concepts and empirical hardware data. Prior to this project, tools like Cinebench and HWiNFO64 are new to me, and observing their real-time telemetry made the intricate behavioral dynamics of the CPU deeply clear.

It was fascinating to witness the complex interdependencies among system component under stress, particularly watching how physical thresholds like thermal limits immediately intervene to scale back clock frequencies. This hands-on experience fundamentally reshaped my perspective on engineering, teaching me that software efficiency and data management are inextricably linked to the underlying physics of the machine.

Finally, this study provided me with an invaluable practical foundation that will undoubtedly guide my approach to programming and hardware optimization in my future study.

4.2.2 Reflection by Akmal Rafique bin Mohd Raphaie (HWiNFO64)

Doing this project was a massive eye opener for me. Going into it, I thought HWiNFO64 was just going to be another basic system monitor. But actually running those heavy Cinebench stress tests on my sister's old laptop really showed me what goes on under the hood. Watching the CPU hit over 90 degrees and seeing it throttle itself almost a thousand times during Workload A made me finally understand why the laptop's cooling fans got so incredibly loud.

Figuring out how to configure the tool with Najmi was a bit hard at first because there are way too many sensors and numbers on the screen. But once we managed to hide all the extra noise and isolate just the core utility and temperatures for Workload B, it was actually really satisfying.

The best part of this whole benchmark was definitely seeing the dynamic thread migration in action. Watching Windows literally bounce the heavy workload from one core to another just to keep the laptop from melting. It really connected the dots between the theoretical stuff we learn in our COA classes and the actual physical hardware sitting right in front of me.

4.2.3 Reflection by Nasaai bin Noriskamar (PassMark PerformanceTest)

Alhamdulillah, through completing this Computer Organization and Architecture (COA) project, I had achieved a great understanding of how theoretical processor design function in real-world hardware. In the lecture, we learned about concepts like registers, execution, and instruction sets only on reading the slide. However, by hands-on implementing Workload A and Workload B using PassMark PerformanceTest, I can clearly visualize how these microarchitectural components interact. Seeing how scalar sorting instructions can stress a pipeline compared to how efficiently parallel AVX2 vector instructions process matrix blocks give me a very solid appreciation of modern CPU design.

Furthermore, this project highlighted the crucial interrelated between hardware software optimization and physical constraints. I have learned that achieving high performance is not just about having a fast clock speed. By observing my laptop's Core i5-1235U balance its temperature at a cool 40.5°C during short bursts, while altering performance scaling across multiple threads, it has taught me on how hardware safety mechanisms operate under actual execution workloads.

In conclusion, this benchmarking assignment has literally improved my technical and analytical skills. Facing challenges like converting raw throughput score into formalized MIPS, CPI, and GFLOPS metrics leading me to redo some revision of our lecture formulas and apply them practically. It has also improved my troubleshooting skills during configuring multi-threaded environments and reading microarchitectural bottlenecks from the system logs.

4.2.4 Reflection by Muhammad Fairuz bin Herman (CrystalDiskMark)

I am happy that I got to learn using these tools as I have never known about these tools before. I have learned something new as I did this project for Computer Organization and Architecture. Seeing how these diagnostics actually monitor system behavior made everything much clearer.

It was interesting to see how a computer handles heavy workloads in real-time. Observing how components depend on each other and how physical limits like heat force a processor to lower its clock speed showed me how the hardware layers are connected.

Ultimately, this project taught me that you cannot write efficient programs or manage data well without respecting the physical limits of the machine. It was a good hands-on experience that gave me a practical foundation for my future studies.

5.0 References

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Appendices

Task Description	Assigned Member
1.0 Introduction & Tool Descriptions	All Members
2.1 Cinebench 2026 Installation & Manual	Najmi
2.2 HWiNFO64 Installation & Manual	Akmal
2.3 PassMark PerformanceTest Installation & Manual	Nasaaie
2.4 CrystalDiskMark Installation & Manual	Fairuz
3.1 Detailed Workload Explanations	All Members
3.2 & 3.3 Execution Flowcharts	All Members
3.4 Analysis and Discussions (Result & Limitations)	All Members
4.0 Conclusion and Technical Reflections	All Members
5.0 Academic References & Compilation	All Members

Photos



Through the process of implementation this project, Our group decided to conduct four meeting at four different places. The first meeting was conducted around our own faculty which is N28 for division task and tools. For the second one was at Mc Donald UTM Skudai, the real purpose of conducting meeting there is for having a clear clarification of our own task to avoid miscommunication or misunderstanding between our group member in order to ensure our project completed correctly and successfully while enjoy eating burgers and ice-cream there. And for the third meeting is around our faculty again at N28a. Lastly, our group decide to conduct a final meeting at PSZ (Perpustakaan Sultanah Zanariah). There we did a last review and finalize our report explicitly before submitting it.